1.1 Evaluating Algebraic Expressions

Name _____ Period ____ Date _____

Algebra I 1.1 · Student sheet

I can…

- evaluate expressions with one variable.
- evaluate expressions with two or more variables using the order of operations.
- write and evaluate expressions for real-life situations.

Vocabulary · fill in the missing word

Variable — a _____ that stands for a number

Algebraic expression — an expression with numbers, variables, and _____

Evaluate — _____ numbers for the variables and simplify

Order of operations — grouping symbols, _____, multiply and divide (left to right), add and subtract (left to right)

Part 1 · Order of operations relay

Evaluate each expression when **x = 3** and **y = −2**. Take turns: one partner does a step, the other checks it and does the next.

Expression	Work	Value
$4x - 5$		
$x^2 + 2y$		
$3(x - y)$		
$(x + y)^2$		
$-y^2 + x$		
$xy - \frac{x}{3}$		

Part 2 · Notes and examples (fill in each note, then work the example)

Order of operations: Order of operations: _____, _____, multiply and _____ (left to right), add and _____ (left to right).

Evaluating: To evaluate, _____ each variable with its value, using _____ around negatives.

Careful with negatives: -3^2 = _____, but $(-3)^2$ = _____.

Example 1

Evaluate each expression.

a. $5x - 8$ when $x = 4$ b. $2x^2 + 1$ when $x = -3$

Example 2

Evaluate $a^2 - 3b$ when $a = 4$ and $b = -2$.

Example 3

Evaluate $\frac{x + y}{z} - 2y$ when $x = 7$, $y = 5$, and $z = 3$.

Example 4

A phone plan costs \$25 a month plus \$0.10 per text. The cost is $C = 25 + 0.1t$, where t is the number of texts. Find the cost for 140 texts.

Part 3 · Partner practice

Partner 1

Evaluate $3x^2 - x$ when $x = -2$.

Partner 2

Evaluate $4m - n^2$ when $m = 3$ and $n = -5$.